

Closing the Loop: Calibration-Aware Planning with Evolving Digital Twins

Research Proposal

Motivation and Literature

Digital twins and Bayesian calibration. A digital twin uses a computer or physics-based model to represent a physical system, but its predictions depend on unknown physical parameters and remain subject to model-form error. Bayesian calibration therefore represents a field response through a calibrated simulator plus discrepancy,

$$y = \eta_\phi(x; \theta) + \delta(x) + \epsilon,$$

where η_ϕ is the computer model, θ contains calibration parameters, and δ captures systematic simulator bias. Our recent Bayesian Recursive Projected Calibration (BRPC) framework (Xu and Park, 2026) extends this formulation to streaming physical data, tracking time-varying calibration parameters while preserving separation between calibration and discrepancy, and incorporating restart mechanisms for abrupt regime changes. However, calibration is treated as an inference problem: the operating decisions that generate the physical observations are not chosen to improve downstream decisions.

Digital twins as world models. For sequential decision making, the digital-twin input naturally decomposes as

$$x_t = (s_t, a_t),$$

where s_t is an observed but non-controllable operating state/context and a_t is a controllable operating action. If the twin predicts the next physical state, then

$$s_{t+1} \approx \eta_\phi(s_t, a_t; \theta_t)$$

is precisely a structured *world model*. Thus, digital-twin calibration and world-model learning describe the same object from two complementary perspectives: θ_t is a calibration parameter from the digital-twin perspective and a latent physical state from the sequential-decision perspective.

Recent work has begun connecting these two perspectives. The Actor–Simulator framework (Zheng et al., 2025) integrates digital-twin calibration with model-based reinforcement learning and uses policy-relevant uncertainty to guide physical experiments. Reinforcement Twinning (Schena et al., 2024) similarly couples system identification of a digital twin with control-policy learning. Emerging work also studies trajectory design explicitly for data-efficient twin calibration (Spitkovska and Kuzmenko, 2026). These approaches establish the importance of coupling calibration and control, but largely focus on identifying an unknown but fixed physical model or explicitly excitation-oriented calibration. They do not address a calibrated twin whose physical parameters continually drift or jump during deployment while the *same physical action* must jointly trade off operational reward and the future value of recalibration.

Time-varying and cost-aware Bayesian optimization. Time-varying BO models a changing black-box response $f_t(x)$ directly, for example through temporal Gaussian-process structure (Bogunovic et al., 2016) or weighted nonstationary posteriors (Deng et al., 2022). Cost-aware and nonmyopic BO instead emphasizes that decisions should account for future evaluations and heterogeneous or movement costs. Lee et al. (Lee et al., 2021) formulate cost-constrained BO as a CMDP, while movement-aware methods explicitly penalize $d(a_{t-1}, a_t)$ (Ramesh et al., 2022). These approaches provide useful sequential-optimization tools, but treat nonstationarity primarily at the black-box response level rather than through an evolving, physically interpretable calibration state of a digital twin. [TODO: coupling of action to calibration; uq aspect, gp bayesian; add r0 into Ht, expectation: no variance uq; quantile: robust control, eta could be expensive, surrogate?]

s'—s,a, p(b=theta,delta), p(theta_{t+1}|theta_t, s, a); s, a—> thetaposterior/informationcostinto value?LCB]

Dual control and nonmyopic planning. Dual control recognizes that an action has both a *control effect* and an *information effect*: it changes the system while also changing what is known about uncertain dynamics. Exact dual control is a belief-space dynamic program and is generally intractable. Sampling-based dual MPC therefore uses scenario trees with Bayesian parameter updates along hypothetical future trajectories (Arcari et al., 2020). Related nonmyopic BO work replaces nested lookahead optimization with jointly optimized one-shot scenario trees (Jiang et al., 2020). These ideas provide a computational foundation for our planner, but do not incorporate continual digital-twin recalibration under structured parameter drift, residual simulator discrepancy, and abrupt changes.

Gap. The open problem is therefore not merely how to calibrate a digital twin, nor how to optimize a given world model. We consider the closed-loop deployment problem in which

actions operate the physical system and simultaneously determine how well its evolving digital twin can be recalibrated.

The objective is to learn only the uncertainty that is valuable for future decisions while maintaining high operational reward and avoiding costly operating changes.

Problem Formulation

Let $s_t \in \mathcal{S}$ denote the observed physical state or operating context, $a_t \in \mathcal{A}$ the controllable action, and $\theta_t \in \Theta \subset \mathbb{R}^p$ an unobserved physical calibration state. A pretrained or physics-based digital twin η_ϕ predicts the physical transition:

$$s_{t+1} = \eta_\phi(s_t, a_t; \theta_t) + \delta_t(s_t, a_t) + \epsilon_t, \tag{1}$$

where δ_t captures irreducible model discrepancy. More generally, the digital-twin output may jointly contain next-state and reward responses,

$$y_t = (s_{t+1}, r_t),$$

so the standard response-level BRPC formulation is recovered as a special case.

The physical regime itself evolves according to a *dynamic calibration prior*

$$\theta_{t+1} \sim p_\psi(\theta_{t+1} \mid \theta_t, s_t, a_t), \quad (2)$$

with gradual evolution within operating regimes and occasional abrupt changes. The simpler exogenous model $p_\psi(\theta_{t+1} \mid \theta_t)$ is included as a special case. Equation (2) allows physical mechanisms such as degradation, aging, or wear to make the future twin predictable without assuming that the full physical dynamics must be relearned online.

Given the physical history

$$\mathcal{H}_t = \{s_0, a_0, s_1, \dots, a_{t-1}, s_t\},$$

an extension of BRPC maintains the calibrated belief

$$b_t = p(\theta_t, \delta_t \mid \mathcal{H}_t). \quad (3)$$

Thus θ_t acts simultaneously as an online calibration parameter and as a latent physical state of the world model.

We seek a policy maximizing operational reward while penalizing costly action changes:

$$J_T^\pi = \mathbb{E}^\pi \left[\sum_{t=0}^{T-1} \gamma^t \{r(s_t, a_t) - \lambda c(a_{t-1}, a_t)\} \right]. \quad (4)$$

The movement term may represent recipe switching, ramping energy, reconfiguration time, or transient operating risk.

Calibration-Aware Planning

Although (4) contains no explicit information bonus, information acquisition arises automatically because the action affects the next calibration belief. Conditional on b_t , the history is summarized by the belief-state

$$(s_t, b_t, a_{t-1}),$$

which yields the finite-horizon Bellman recursion

$$\begin{aligned} V_H(s, b, a^-) = \max_{a \in \mathcal{A}} \mathbb{E} \Big[& r(s, a) - \lambda c(a^-, a) \\ & + \gamma V_{H-1}(s', \mathcal{U}_{\text{BRPC}}(b; s, a, s'), a) \Big]. \end{aligned} \quad (5)$$

Here s' is sampled from the calibrated digital twin (1), and

$$b' = \mathcal{U}_{\text{BRPC}}(b; s, a, s')$$

is the posterior that would result from observing that hypothetical physical transition.

Equation (5) reveals the key dual effect:

$$a_t \longrightarrow s_{t+1} \longrightarrow b_{t+1} \longrightarrow \text{future actions}.$$

An action may therefore be optimal even when its immediate reward is lower, provided that its physical response is sufficiently informative to distinguish plausible calibration states and materially improve future decisions. Importantly, this is *decision-relevant* information acquisition: uncertainty is valuable only when resolving it changes downstream control.

A simple UCB-style rule,

$$a_t^{\text{UCB}} = \arg \max_a \{ \mu_t^r(s_t, a) + \beta_t \sigma_t^r(s_t, a) - \lambda c(a_{t-1}, a) \},$$

is a useful computational baseline but remains myopic: it rewards uncertainty itself rather than the downstream value of resolving uncertainty. Likewise, certainty-equivalent MPC replaces b_t by a posterior mean and ignores the effect of a_t on future calibration.

Proposed planner. We will approximate (5) using a short-horizon, posterior-contingent scenario tree. For each candidate action, we:

1. sample (θ_t, δ_t) from b_t and simulate a future physical response with the digital twin;
2. perform a *fantasy BRPC update* to obtain the branch-specific posterior b_{t+1} ;
3. optimize the next action separately under each resulting belief;
4. recursively continue for a short horizon H and execute only the root action before recalibrating with the real observation.

Following one-shot multi-step optimization (Jiang et al., 2020), actions at all scenario-tree nodes can be optimized jointly rather than through deeply nested maximizations. A two-step version provides a lightweight decision-value-of-calibration-information approximation.

Expected Contributions and Theory

C1: An evolving calibrated digital twin as a world model. We formulate sequential digital-twin deployment by decomposing the physical transition into stable simulator structure η_ϕ , an evolving low-dimensional calibration state θ_t , and residual discrepancy δ_t . This makes BRPC-style calibration directly usable as Bayesian state estimation for world-model planning.

C2: Closing the calibration–decision loop. We develop a planner in which the same physical action simultaneously affects immediate reward, future physical states, switching cost, and the quality of future digital-twin calibration. This differs from separately designing calibration experiments and control actions.

C3: Finite-time guarantees linking calibration to control. The target theoretical path consists of three steps.

T1: Calibration error $\Rightarrow$ transition-model error. If η_ϕ is L_θ -Lipschitz in θ , establish

$$\sup_{s,a} W_1\left(P_t(\cdot | s, a), \hat{P}_t(\cdot | s, a)\right) \leq L_\theta e_t^\theta + e_t^\delta + e^\eta \equiv \epsilon_t^{\text{twin}},$$

where e_t^θ follows from BRPC tracking, e_t^δ from discrepancy estimation, and e^η represents irreducible simulator/emulator error. The proof couples true and calibrated transitions using common process noise, then applies Lipschitz continuity and the BRPC tracking guarantee.

T2: Twin error $\Rightarrow$ planning-value error. For bounded Lipschitz rewards, use Bellman induction and Kantorovich–Rubinstein duality to show

$$\left| V_H^\pi(s, b; P) - V_H^\pi(s, b; \hat{P}) \right| \leq \sum_{h=0}^{H-1} C_{H,h} \epsilon_{t+h}^{\text{twin}}.$$

T3: End-to-end performance under drift and abrupt changes. Against an oracle receding-horizon controller using the true evolving twin, target

$$J_T^{\text{oracle}} - J_T^{\text{alg}} \lesssim C_H \sum_{t=1}^T \epsilon_t^{\text{twin}} + \sum_{t=1}^T \epsilon_t^{\text{plan}} + C_{\text{jump}} \sum_{j=1}^K D_j + C_{\text{FA}} N_{\text{FA}},$$

where ϵ_t^{plan} is scenario-tree approximation error, D_j is change-detection delay, and N_{FA} is the number of false restarts.

Finally, a two-regime separation result will construct an instance where certainty-equivalent or myopic UCB control repeatedly selects low-information actions, whereas the belief-space planner accepts a small immediate reward loss to identify the current calibration regime and achieves strictly larger multi-step return. This theorem will isolate the value of *calibration-aware planning* from the value of BRPC calibration itself.

References

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